

SUSHMA UPADHAYAY

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EDUCATION

Bucknell University - B.S. Computer Science & Data Science

Aug 2025 - May 2027

- GPA: 3.56/4.0 | Dean's List

Coursework: SQL, AI and Cloud Computing, Applied Machine Learning for Business, Data Science Fundamentals, Data Structures & Algorithms, Database Management, Software Engineering and Design, Programming Language Design

Honors: American Statistical Association Scholar, Tri Alpha Honor Society, National Society of Leadership and Success, Bucknell Summer Scholar (2024)

Harrisburg Area Community College - A.S. Mathematics and Computer Science

May 2022 - Dec 2024

- GPA: 3.86/4.0, Highest Honors, Dean's List | Phi Theta Kappa Honor Society

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C, C++, Bash, JavaScript, HTML/CSS

Data Science & ML: Pandas, NumPy, scikit-learn, statsmodels, Hugging Face Transformers (BERT, RoBERTa), large language models, natural language processing, logistic regression, random forest, fairness metrics (equalized odds, calibration), exploratory data analysis, statistical modeling

Tools & Platforms: Git/GitHub, Jupyter, Tableau, RStudio, VS Code, PyCharm, IntelliJ, Vim

EXPERIENCE

Dominguez Center for Data Science, Bucknell University

Senior Data Science Student Fellow

Jan 2026 - Present

- Lead a 2-person research team building an AI-assisted, human-reviewed workflow for coding open-ended student survey responses (NSSE, Getting to Know You); set the project timeline, review teammates' work, and report progress to project mentors and the stakeholder.
- Apply large language models to label response themes and Hugging Face transformers (BERT, RoBERTa) for aspect-based sentiment, then measure how closely AI labels agree with human labels; deliver results as a Tableau dashboard, an institutional summary report, and a documented code repository.

Data Science Fellow

- Worked on a 3-person team, mentored by the CEO of ShiftMed (a nurse staffing app), to analyze the app's data and find what drives retention and how the most active users perform.
- Built dashboards and presented our findings to ShiftMed stakeholders and a non-technical audience.

Bucknell University, Department of Computer Science

Undergraduate Researcher, Machine Learning to Analyze Algorithmic Fairness

Jan 2026 - Present

- Built an end-to-end machine learning pipeline in Python (Pandas, NumPy, scikit-learn) to reconstruct the PACT and STRONG-R risk assessment tools used by the Pennsylvania Department of Corrections (PADOC), using random forest and logistic regression to reach 92% predictive accuracy.
- Designed a fairness evaluation framework measuring predictive disparities across race, gender, and age, producing a systematic audit of bias in state correctional risk models.

GorkhaTech (Remote)

Software Developer Intern

Aug 2024 - Feb 2025

- Built the Bright Community of Central Pennsylvania (BC2P) web platform with C#, JavaScript, HTML, and CSS, making local job listings easier for community members to search and access.
- Worked in an agile team to reproduce, troubleshoot, and resolve reported issues, shortening the turnaround time on bug fixes across sprints.

LEADERSHIP AND ACTIVITIES

Secretary, Society of Women Engineers

Jan 2026 - Present

- Manage organization-wide communication and records for 20+ members, keeping meeting notes, rosters, and announcements accurate and easy to find.

Vice President of Scholarship, Phi Theta Kappa Honor Society

May 2024 - Jan 2025

- Partnered with chapter officers and faculty to run honors study workshops and campus events, coordinating logistics and turnout.

Member: Computer Science Club, Machine Learning Club, Entrepreneurship & Innovation Club, South Asian Engineers Club